

HashSet

Managing Unique Collections in Java

codehive

codehive

Uniqueness Simplified

A HashSet is a collection that **cannot contain duplicate elements**. It is part of the `java.util` package and implements the `Set` interface. It uses a mechanism called "Hashing" to store data efficiently.

```
import java.util.HashSet;

public class Main {
    public static void main(String[] args) {
        HashSet<String> cars = new HashSet<>();
        cars.add("Volvo");
        cars.add("BMW");
        cars.add("Ford");
        cars.add("BMW"); // Duplicate - ignored!
        cars.add("Mazda");
        System.out.println(cars);
    }
}
```

Core Characteristics

1. No Duplicates: Every item in a HashSet must be unique. If you try to add an item that already exists, the Set remains unchanged.

2. Unordered: Unlike an ArrayList, a HashSet does not maintain the order of insertion. When you print a HashSet, the items may appear in a random-looking sequence.

3. High Performance: HashSet offers constant-time performance ($O(1)$) for basic operations like add, remove, and contains, making it much faster for searching than a List.

Common Methods

- `cars.contains("BMW")`: Checks if an item exists.
- `cars.remove("Volvo")`: Removes an item.
- `cars.clear()`: Removes all items.
- `cars.size()`: Returns the number of unique items.

Collection Frameworks with

codehive

Mastering data uniqueness.

codehive